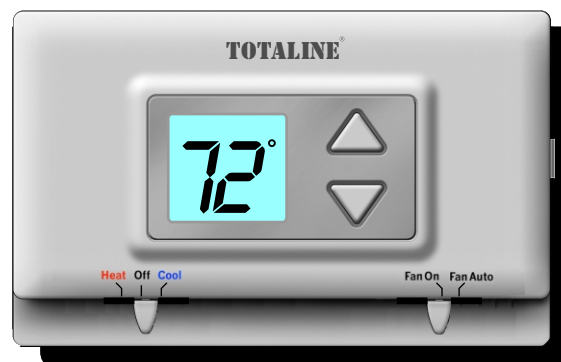


OWNER'S MANUAL

P/N P474-0130

Non-Programmable Digital Thermostat



- Millivolt Compatible • DC Voltage Compatible
- Battery Operated • Stages: 1-Heat, 1-Cool
- Digital Display • Easy Operation

TOTALINE®

Star

Use with most Air Conditioning & Heating Systems including: 1 Stage
Electric Cooling & Gas Heating, Heat Pump, Electric Heat, Hydronic Heat,
Fan Coils and PTAC units..

Replacement Components Division - Carrier Corporation - Made in China

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INSTALLATION INSTRUCTIONS

Safety Warnings

P/N P474-0130

CAUTION Follow *Installation Instructions* carefully.



DISCONNECT POWER TO THE HEATER -
AIR CONDITIONER **BEFORE** REMOVING
THE OLD THERMOSTAT AND INSTALLING
THE NEW THERMOSTAT.



WARNING



CAUTION

The 2 Alkaline "AA" batteries must be replaced at least every 12 months to assure proper operation. The thermostat will display the Low Battery code (fig. 1) on the display of the thermostat when it is time to replace the batteries.



FIG. 1

When **Lb** is displayed the batteries must be replaced immediately. The manufacturer cannot be liable for improper operation of the thermostat if the batteries are not immediately replaced.

The annual battery replacement is especially critical in locations subject to freezing temperatures. The thermostat will be unable to turn on the Heat if the batteries are exhausted.

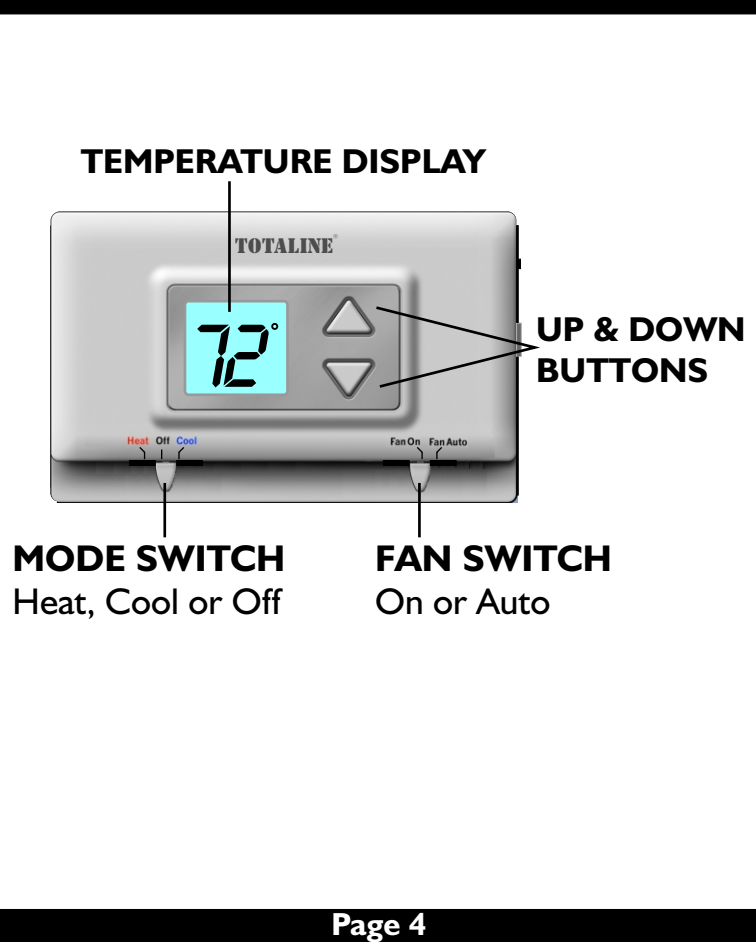
This device complies with Part 15 of the FCC rules.

Operation is subject to the following 2 conditions:

- (1) This device may not cause harmful interference, and (2)
- This device must accept any interference received, including interference that may cause undesired operation.

Replacement Components Division • Carrier Corporation • Patents Pending

Location of Controls



Display

①



②



①

Current room temperature.

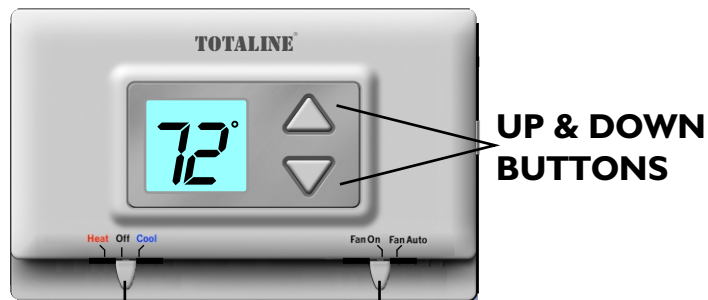
②

If the Up or Down arrow buttons are pressed the thermostat will show the desired Set Temp temperature indicator. Once this screen is reached you may use the Up or Down arrow buttons to adjust the desired room temperature.

③

After five seconds with no button presses the thermostat will revert back to show the current room temperature.

Normal Operation

**MODE SWITCH**

Heat, Cool or Off

FAN SWITCH

On or Auto

Manual Operation

- ❶ Select Heat or Cool with the mode switch.
- ❷ Normally leave the fan switched to Fan Auto.
In Fan Auto, the fan will turn on only with a heat or cool demand. When Fan On is selected, the fan will run continuously, even when the mode switch is set to Off.
- ❸ Adjust the desired set temperature with the Up or Down buttons.

INSTALLATION INSTRUCTIONS

Step #1

Preparation



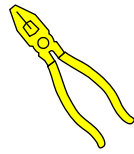
Proper installation of the thermostat will be accomplished by following these step by step instructions. If you are unsure about any of these steps, call a qualified technician for assistance.



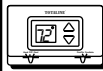
These tools will be required:



*Flat Blade
Screwdriver*



*Wire cutter
& Stripper*



Make sure your Heater/Air Conditioner is working properly before beginning installation of the thermostat.



Carefully unpack the thermostat. Save the screws and instructions.

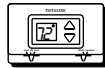


Turn off the power to the Heating/Air Conditioning system at the main fuse panel. Most residential systems have a separate breaker for disconnecting power to the furnace.

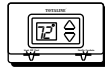
INSTALLATION INSTRUCTIONS

Step #2

Remove & Replace Old Thermostat



Remove the cover of the old thermostat. If it does not come off easily check for screws.



Loosen the screws holding the thermostat base or subbase to the wall, and lift away.



Disconnect the wires from the old thermostat. Tape the ends of the wires as you disconnect them, and mark them with the letter of the terminal for easy reconnection to the new thermostat.



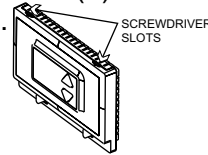
Keep the old thermostat for reference purposes, until your new thermostat is functioning properly.

INSTALLATION INSTRUCTIONS

Step #3 Installation / Battery Replacement

Open The New Thermostat

- ① The top of the thermostat housing has two (2) screw-driver slots to assist when separating.
- ② To pull the housing apart, insert a small blade screw-driver into the slot and rotate 90°. This will release the top housing snaps.
- ③ Repeat the procedure in the other screw driver slot.
- ④ Separate the housing halves by pulling the top forward until the pins release, and then lift the bottom out.



The batteries must be replaced immediately when the thermostat displays the Low Battery code (fig.1).



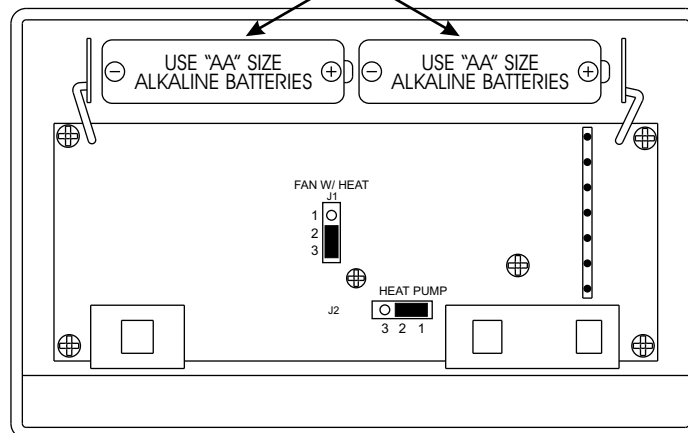
FIG. 1

INSTALLATION INSTRUCTIONS

Battery Replacement

- 5** REPLACE WITH ALKALINE BATTERIES AT LEAST ONCE EVERY YEAR, OR WHEN THE “LOW BATTERY”  ICON APPEARS (pages 3,9).

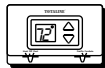
POSITION BATTERIES AS SHOWN



INSTALLATION INSTRUCTIONS

Step #4

Wire Connections



If the terminal designations on your old thermostat do not match those on the new thermostat, **refer to the chart below, or the wiring diagrams that follow.**

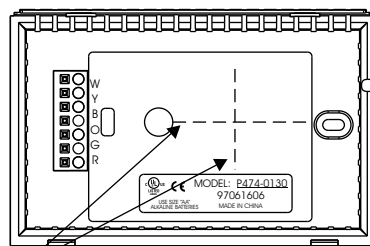
Wire from the old thermostat terminal marked	Function	Install on the new thermostat connector marked
W1, W or H	Heating	W
Y1 or Y	Cooling	Y
B	Rev. Valve (Energize to Heat)	B
O	Rev. Valve (Energize to Cool)	O
G or F	Fan	G
Rh, R, M, Vr, A	Power	R

Thermal Insulating Sheet

A label is provided on the backplate that prevents drafts, originating inside the wall, from entering the thermostat.

These drafts, left unchecked, may cause incorrect room temperature readings.

Please do not remove this label from the thermostat. Insert the wires through the slots provided in the label as shown in Fig. 1



Wire Slots

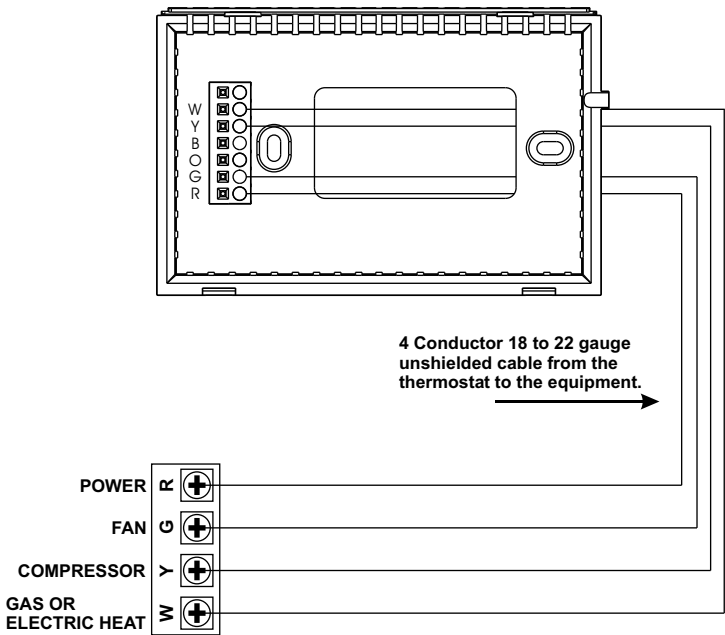
Fig. 1

INSTALLATION INSTRUCTIONS

Sample Wiring Diagrams
Gas or Electric Heat

4 Wire, 1 Stage Cooling, 1 Stage Gas Heat

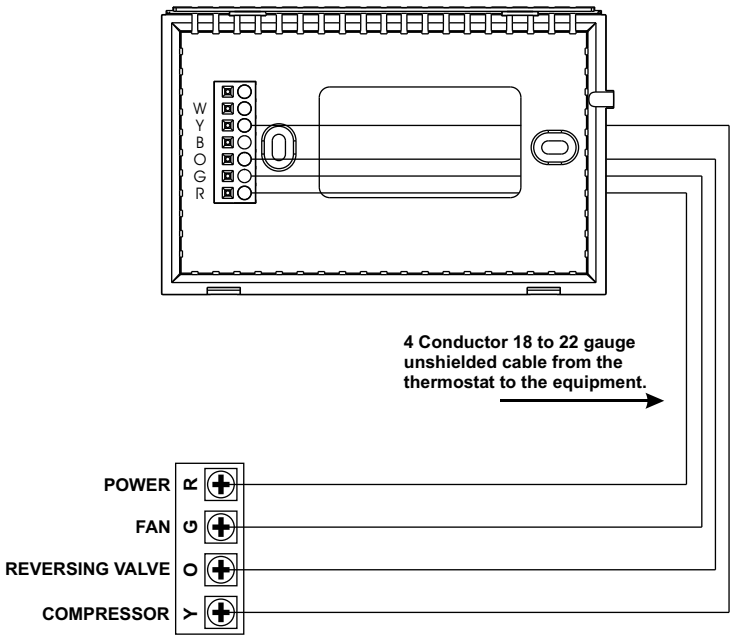
Residential Gas or Electric Heat *,
Electric Cool, split systems & package
units



INSTALLATION INSTRUCTIONS

Sample Wiring Diagrams
Gas or Electric Heat

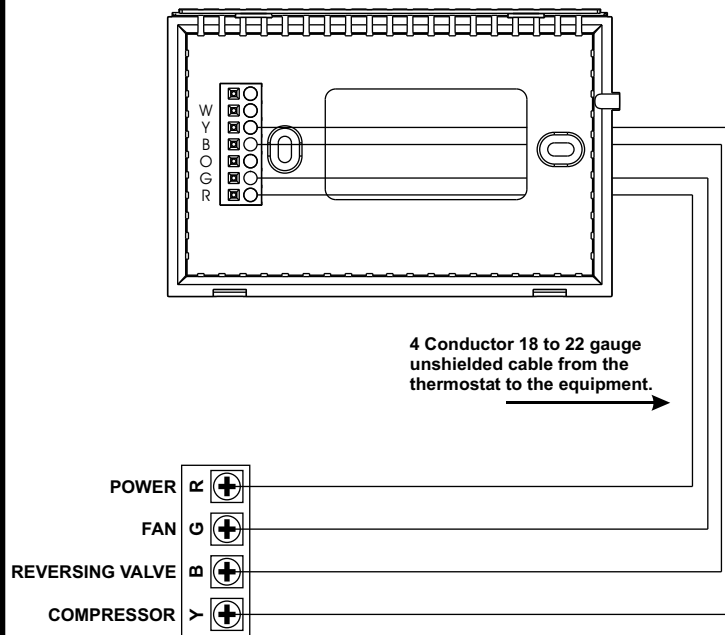
4 Wire, 1 Stage Cooling, 1 Stage Heat-Heat Pump with O reversing valve.
Residential Heat Pumps, split systems & package units, with no auxiliary heat.



INSTALLATION INSTRUCTIONS

Sample Wiring Diagrams Gas or Electric Heat

4 Wire, 1 Stage Cooling, 1 Stage Heat-Heat Pump with B reversing valve.
Residential Heat Pumps, split systems & package units, with no auxiliary heat.

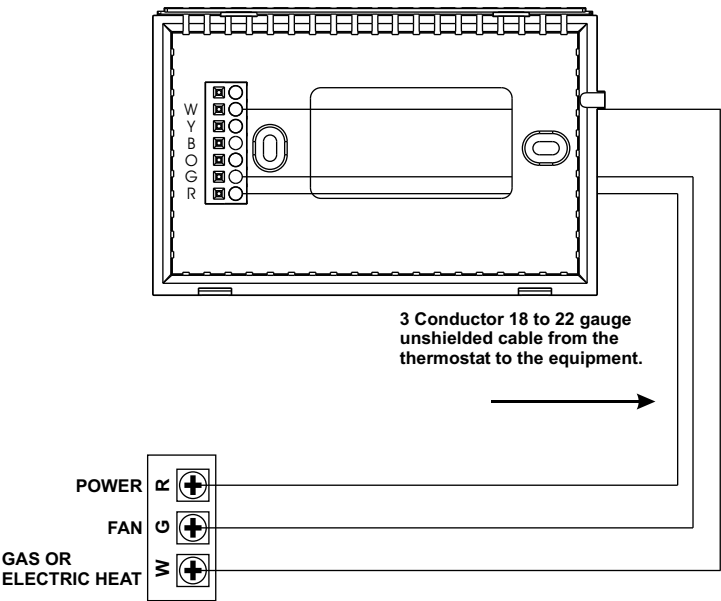


INSTALLATION INSTRUCTIONS

Sample Wiring Diagrams
Gas or Electric Heat

3 Wire, 1 Stage Heat

Residential Gas or Electric Heat units
with a separately controlled fan.

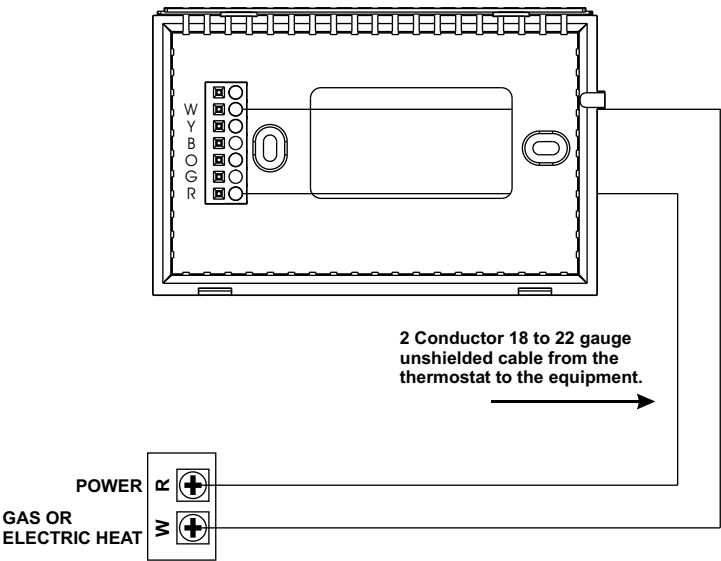


INSTALLATION INSTRUCTIONS

Sample Wiring Diagrams
Gas or Electric Heat

2 Wire, 1 Stage Gas Heat

Residential Gas or Millivolt units.

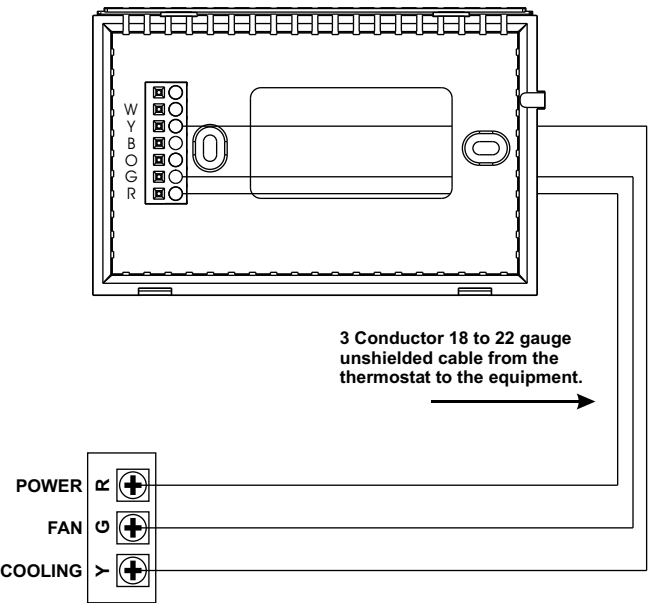


INSTALLATION INSTRUCTIONS

Sample Wiring Diagrams
Gas or Electric Heat

3 Wire, 1 Stage Cooling

Residential Electric Cool units

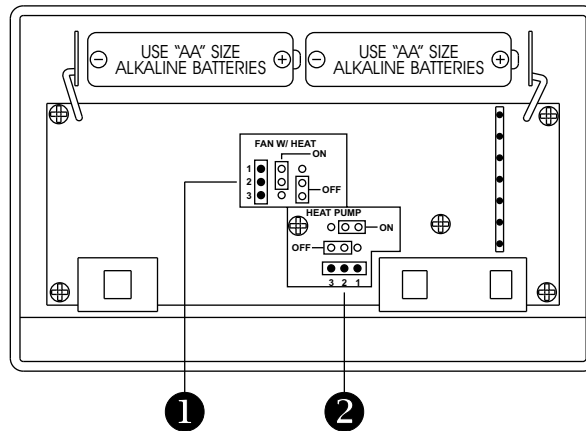


INSTALLATION INSTRUCTIONS

Step #5

Jumper Configuration

REPLACE WITH ALKALINE
BATTERIES ONCE EVERY YEAR



If the HVAC unit has First Stage Electric Heat then jumper ❶ should be set to ON. If the jumper is set for Fan w/ Heat On, the fan will energize immediately on a call for heating. For all other applications this jumper should remain set to the OFF position.



If the HVAC unit is a Heat Pump, then jumper ❷ should be set to ON. For all other applications this jumper should remain set to the OFF position.

INSTALLATION INSTRUCTIONS

Step #6

Test Operation



Turn on the power to the Heating/Air Conditioning system.



Adjust the Slide Switch until it is located under the word **HEAT** on the thermostat. Press the Up or Down buttons until the set temperature is 10 degrees above room temperature. The HVAC unit should energize in the heating mode (pages 5-6).



Adjust the Slide Switch until it is located under the word **COOL** on the thermostat. Press the Up or Down buttons until the set temperature is 10 degrees below room temperature. The HVAC unit should energize in the cooling mode (pages 5-6).



Adjust the Slide Switch until it is located under the word **OFF**. Adjust the other slide switch until it is located under the word **Fan On**. The fan should turn on and run continuously (pages 5-6).

INSTALLATION INSTRUCTIONS

Trouble Shooting



SYMPTOM: The slide switches on the thermostat are very difficult to move.

CAUSE: The backplate of the thermostat is deformed by being screwed tightly into a wall that is not perfectly flat.

REMEDY: Loosen the screws holding the thermostat into the wall.



SYMPTOM: The air conditioning does not attempt to turn on.

CAUSE: The cooling setpoint is set too high or the Mode Switch is not set for Cool, or the batteries are too weak.

REMEDY: Consult the Normal Operation section of this manual to lower the cooling setpoint and to correct the Mode Switch position, or replace the batteries (page 10).

INSTALLATION INSTRUCTIONS

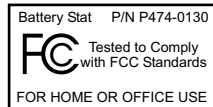
Trouble Shooting



SYMPTOM: The heating does not attempt to turn on.

CAUSE: The heating setpoint is set too low or the Mode Switch is not set for Heat, or the batteries are too weak.

REMEDY: Consult the Normal Operation section in this manual to raise the heating setpoint and to correct the Mode Switch position (page 10), or replace the batteries.



P/N 88-396
Rev. 2

INSTALLATION INSTRUCTIONS

Warranty

One-Year Warranty - This Product is warranted to be free from defects in material and workmanship. If it appears within one year from the date of original installation, whether or not actual use begins on that date, that the product does not meet this warranty, a new or remanufactured part, at the manufacturer's sole option, to replace any defective part will be provided without charge for the part itself; PROVIDED the defective part is returned to the distributor through a qualified servicing dealer.

THIS WARRANTY DOES NOT INCLUDE LABOR OR OTHER COSTS incurred for diagnosing, repairing, removing, installing, shipping, servicing or handling of either defective parts or replacement parts. Such costs may be covered by a separate warranty provided by the installer.

THIS WARRANTY APPLIES ONLY TO PRODUCTS IN THEIR ORIGINAL INSTALLATION LOCATION AND BECOMES VOID UPON REINSTALLATION.

LIMITATIONS OF WARRANTIES – ALL IMPLIED WARRANTIES (INCLUDING IMPLIED WARRANTIES OF FITNESS FOR A PARTICULAR PURPOSE AND MERCHANTABILITY) ARE HEREBY LIMITED IN DURATION TO THE PERIOD FOR WHICH THE LIMITED WARRANTY IS GIVEN. SOME STATES DO NOT ALLOW LIMITATIONS ON HOW LONG AN IMPLIED WARRANTY LASTS, SO THE ABOVE MAY NOT APPLY TO YOU. THE EXPRESSED WARRANTIES MADE IN THIS WARRANTY ARE EXCLUSIVE AND MANY NOT BE ALTERED, ENLARGED, OR CHANGED BY ANY DISTRIBUTOR, DEALER, OR OTHER PERSON WHATSOEVER. ALL WORK UNDER THE TERMS OF THIS WARRANTY SHALL BE PERFORMED DURING NORMAL WORKING HOURS. ALL REPLACEMENT PARTS, WHETHER NEW OR REMANUFACTURED, ASSUME AS THEIR WARRANTY PERIOD ONLY THE REMAINING TIME PERIOD OF THIS WARRANTY.

THE MANUFACTURER WILL NOT BE RESPONSIBLE FOR:

1. Normal maintenance as outlined in the installation and servicing instructions or owners manual including filter cleaning and/or replacement and lubrication.
2. Damage or repairs required as a consequence of faulty installation, misapplication, abuse, improper servicing, unauthorized alteration or improper operation.
3. Failure to start due to voltage conditions, blown fuses, open circuit breakers or other damages due to the inadequacy or interruption of electrical service.
4. Damage as a result of floods, winds, fires, lightning, accidents, corrosive environments or other conditions beyond the control of the Manufacturer.
5. Parts not supplied or designated by the Manufacturer, or damages resulting from their use.
6. Manufacturer products installed outside the continental U.S.A., Alaska, Hawaii, and Canada.
7. Electricity or fuel costs or increases in electricity or fuel costs from any reason whatsoever including additional or unusual use of supplemental electric heat.
8. ANY SPECIAL INDIRECT OR CONSEQUENTIAL PROPERTY OR COMMERCIAL DAMAGE OF ANY NATURE WHATSOEVER. Some states do not allow the exclusion of incidental or consequential damages, so the above may not apply to you.

This warranty gives you specific legal rights, and you may also have other rights which may vary from state to state.